Stellenbosch University Faculty of Engineering

Module Framework

This document should be read with the following documents:

- ***Stellenbosch University Calendar Parts 1 and 11.***
- ***Faculty of Engineering Assessment Rules***
- ***Faculty of Engineering General Stipulations for Undergraduate Modules[1]***

Design (E) 314 2026 (46833-314)	Lecturer(s): Prof Herman Engelbrecht, E4019, hebrecht@sun.ac.za Dr Armand du Plessis, E3016, armandd@sun.ac.za Internal moderator: Prof Callen Fisher	Approved by Programme Coordinator: Date: 2026/02/09

1 Assessment Details

- The dates, venues (if applicable) and any additional information for the main assessments will be provided via learn.sun.ac.za
- Method of assessment is **project based** with assessment components, mark weights, and details as indicated below.
- Note that awarding a pass mark is subject to meeting each ECSA Graduate Attribute assessed in this module, as stated in Faculty of Engineering's Assessment Rules.

Calculation of final marks (according to formulas in the Faculty of Engineering's Assessment Rules):

Division	Full Marks	Sub-minimum
Tests (T)	T1(10%) + T2(20%) + T3(20%) + T4(50%) = 100%	50%
Demonstrations (D)	D1(15%) + D2(15%) + D3(40%) + D4(30%) = 100%	50%
Report (R)	PR(15%) + FR(85%) = 100%	50%
Performance Mark	PM = (T + D + R) / 3	
	PM_{max} = minimum(T, D, R) + 15	

Examples:

T = 65.7%	D = 80%	R = 80%	=> PM = 75% (distinction)
T = 50%	D = 80%	R = 80%	=> PM = 65% (pass, limited by PM_{max})
T = 48.6%	D = 80%	R = 80%	=> PM = 45% (fail, sub-minimum not achieved)

Assessment format:

- All assessment dates and deadlines are indicated in the **module schedule** (section 5) of this module framework.
- Test assessments (T1, T2, T3 and T4) will be invigilated sit-down **open-book** assessments on **STEMLearn** (Venues to be confirmed).
- Demonstration assessments (D1, D2, D3, and D4) will be face-to-face in the undergraduate laboratories using our **automated test stations** at your allotted timeslots.
- The **Preliminary Report (PR)** will be peer reviewed using STEMLearn and a pre-defined rubric.
- The final **Technical Project Final Report (FR)** will be electronically submitted using STEMLearn (and Turnitin) and assessed manually by the lecturers according to a pre-defined rubric. A summarised version of the rubric will be supplied to the students well in advance.
- **The four Graduate Attributes (GA) associated with this module will be assessed in the Final Report (FR) and Preliminary Report (PR).**
- **As this module is using the project-based assessment method:**
  - **NO Dean's Concession Assessment (DCA) is possible or will be allowed if a GA is not attained or the report or demo subminimum is not met.**
  - **If ONLY the test component is required to pass the module, a DCA can be applied for at the end of the year.**

[1] Available on STEMLearn for modules offered by Faculty of Engineering, in the block titled "General Programme Information" on the side of the screen

2 Notional Hours

- You should spend 10 notional hours per credit on this module over the course of the semester.
- It is envisaged that these hours will be allocated as follows:

Activity	Contact Hours	Self-study Hours
Lectures	12 x 0.83h (50 minutes each) = 10	10
Tutorials	-	-
Practicals	36	18
Assignments	4	42
Main assessments	6	30
Total (for this 15-credit module)	**156**	

3 Language of Tuition

The language of tuition in this module is according to the Faculty's approved Language Implementation Plan. Please refer to the website of the Engineering Faculty or the "General Information" block on STEMLearn for the particulars.

4 Module Objectives

Aims:
In this module students will be given the opportunity to develop a small microcontroller based system. Due to logistical constraints certain components have been pre-selected by the lecturer(s). The lecturer(s) will also act as "client" and will determine the user requirements.
Students must interpret the requirements and develop a marketable product. Development comprises: Transforming the user requirements to technical specifications, co-design of the hardware and software, datasheet interpretation, detail circuit design and coding, debugging of hardware and software, measurements, testing and writing a user manual and technical project report.

A student who has successfully completed this module can:
- With guidance, do a small microcontroller-system hardware and software design project containing various digital interfaced external peripherals.
- Use engineering tools and methods to design, implement, debug and measure the system.
- Physically demonstrate the functionality of the developed system using a PC-based test interface.
- With guidance, produce an electronic Technical Project Report to document the development process and results.

Prior knowledge required:
- This module is part of the Computer Systems module chain. It relies on knowledge obtained in Computer Programming 143, Computer Systems 214 and 245, and Electrotechnique 143.

Proceeding application:
- The knowledge developed in this module can be applied further, for E&E students in Design (E) 344 (Project, hardware design, reporting), for E&E and Data engineering students in Project (E) 448 (Project, hardware and software design, reporting), for Mechatronic students Mechatronic Project 478/488 (Project, hardware and software design, reporting). This module is the end of the Computer Systems module chain for Mechatronic program. It finds application in engineering practice areas such as computer-based control, embedded systems, project management and design.

5 Module Content and Schedule

Prescribed textbook(s) and other resources (electronic versions available on stemlearn.sun.ac.za):

1) *Discovering the STM32 Microcontroller, Geoffrey Brown, June 2016.*
2) *STM: STM32F411RE Datasheet – DS10314 Rev 8, UM1724 – Rev 17, RM0383 – Rev 4, PM0214 – Rev 19, ES0287 – Rev 6.*
3) *Accessing Hardware Registers using C, and Selected slides from CS245 – L Visagie/W Smit/W Duckitt.*
4) *Embedded C Coding Standard.*
5) Other component datasheets and application notes as needed.

Content presentation = In person

The **Q&A session,** on Tuesdays at **08:00, for M&M** and **09:00 for E&E (which includes Data Engineering)**, will be facilitated by either one or both lecturers.
All **contact sessions/assignments** will be **Face-to-Face in the Volt and Ampere Labs (E&E 2nd + 4th floor)**.

Week	Lecture Topics	Contact Session/Assignments	Mechatronic	E&E, Data
1 (9 Feb – 13 Feb)	10 Feb – Module overview, Project definition, Project planning, Power supply. Design for testing, STM32F411RE processor, IDE, LEDs, UART Output, Software, Project planning.	Concept Design, Power supply, Nucleo-64, UART Output, Power LED, Status LEDs GPIO Buttons	12 Feb	13 Feb
2 (16 Feb – 20 Feb)	17 Feb – GIT, GPIO Buttons, Debouncing, Interrupts, Temperature sensor, UART Telemetry Request + Command interface, Reporting, Q&A session	Temperature sensor	19 Feb	20 Feb
3 (23 Feb – 27 Feb)	24 Feb – Calibration, PWM, Verification testing, Q&A session, Preparing for a Demonstration. Ultrasonic Sensor (for Demo 2)	**DEMONSTRATION 1**	21 Feb	23 Feb
4 (2 Mar – 6 Mar)	**3 Mar – TEST 1**	UART Telemetry Request, Ultrasonic sensor	5 Mar	6 Mar
5 (9 Mar – 13 Mar)	10 Mar – UART Telemetry Request, Software timing, Verification testing, Q&A session, Light sensor, Op-Amp, ADC concepts (for Demo 3)	**DEMONSTRATION 2**	12 Mar	13 Mar
6 (16 Mar – 20 Mar)	**17 Mar – TEST 2**		-	-
23 Mar – 27 Mar	Test week	-		
28 Mar – 6 Apr	Recess	-		
7 (7 Apr – 10 Apr)	**7 Apr** – Keypad interface, LCD, UART Command interface + handling state machine **@12:00 PRELIMINARY REPORT SUBMISSION (Peer review deadline 13 Apr)**	Keypad, LCD, Command handling state machines, Light sensor, Op-Amp, ADC	9 Apr	10 Apr
8 (13 Apr – 17 Apr)	14 Apr – Menu Interface, SD Card, Data Logging, SPI	UART Command, Menu interface, RTC	Own time	Own time
9 (20 Apr – 24 Apr)	21 Apr – RTC, UART Command, I2C, Accelerometer measurement, Final Syst. Integration	**DEMONSTRATION 3**	23 Apr	24 Apr
10 (27 Apr – 1 May)	28 Apr (Tue) – No lecture, Monday timetable **(Guide for self-learning)**	Accelerometer, SD Card, Data Logging	Own time	Own time
11 (4 May – 8 May)	**5 May – TEST 3**	GPS, System testing	7 May	8 May
12 (11 May – 15 May)	12 May – Reporting, Q&A session	**DEMONSTRATION 4**	14 May	15 May
18 May	18 May @12:00 – **FINAL REPORT SUBMISSION**	-	-	-
9 June	9 June @ 09:00 – **TEST 4**	-	-	-

6 ECSA Knowledge Area Credits

Mathematical Sciences	Natural Sciences	Engineering Sciences	Design and Synthesis	Complementary Studies
0	0	0	11	4
Design and Synthesis: Being a microprocessor based digital design project, most of the content is aimed at assisting the design process. The design of multiple subsystems, culminate in a complete system design and implementation. Software is synthesized to implement the required system functionality according to specifications. Complementary Studies: Through examples and guidance, small scale project management, technical project report writing, and information gathering is learned				

7 ECSA Graduate Attribute(s)

All students are required to complete the same design of a small, embedded computer system, structured as a sequential completion of small sub-systems and functionalities. The project is selected by the lecturer(s) to be of sufficient scope to enable achievement of the following outcomes by the student.

Graduate Attributes GA 3-5 and 9 are assessed in this module. The GAs are assessed for each student individually as described in the table below. The GA assessment shall be based on the student's final report (R) as assessed by the lecturer. A student shall only be eligible for GA assessment if they met the subminima for both the demonstrations (D) and the tests (T). The GA assessment rubric in the appendix shall be completed for each student.

Complex engineering problems

a. These require a fundamentals-based, first principles analytical approach, use in-depth engineering knowledge and have one or more of the following characteristics:

i. A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

ii. Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much of which is at the forefront of the discipline.

iii. Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.

iv. Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.

v. Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

b. and have some or all of the following characteristics:

i. Involve wide-ranging and/or conflicting technical, non-technical issues (such as ethical, sustainability, legal, political, economic, societal) and consideration of future requirements.

ii. Have no obvious solution and require abstract thinking, creativity and originality in analysis to formulate suitable models.

iii. Involve infrequently encountered issues or novel problems.

iv. Address problems not encompassed by standards and codes of practice for professional engineering.

v. Involve collaboration across engineering disciplines, other fields, and/or diverse groups of stakeholders with widely varying needs

vi. Address high-level problems with many components or sub-problems that may require a systems approach.

GA 3. Engineering design: Design creative solutions for complex engineering problems and design systems, components or processes to meet identified needs.

<table>
<tr><td>How is the Attribute Assessed?</td><td colspan="4">The student will perform a procedural design, guided by the lecturer. Planning is only required on a short-term basis. Alternatives are limited by the lecturer. The GA assessment shall be based on the student's preliminary (PR) and/or final report (FR) as assessed by the lecturer. Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon and similar concepts will be evaluated in the reporting section (PR), (FR) of this module.
To be eligible for GA assessment, students will individually perform a set of demonstrations of the subsystems and a final demonstration of the complete system. Each demonstration is followed by a test to verify the student's knowledge of the components, design principles and procedures. Average marks are computed for both demonstrations and tests.</td></tr>
<tr><td rowspan="3">What is Satisfactory Performance?</td><td colspan="4">Using the assessment material and opportunities, the student must show that he/she has performed design and synthesis of components or systems at a level consistent to that which a graduate would participate in an employment situation shortly after graduation. The student must show that he/she:
• Identified and formulated the design problem to satisfy user needs, and applicable standards;
• Planned and managed the design process – focusing on important issues, while recognising and dealing with constraints;
• Acquired and evaluated the requisite knowledge, information and resources, applied correct principles, and used design tools;
• Performed design tasks including analysis, quantitative modelling and optimisation;
• Evaluated alternatives and preferred solutions, exercised judgment, and tested implementation ability;
• Communicated the design logic and information.
In order to achieve the outcome, the student's preliminary (PR) and/or final report (FR) is evaluated, as a whole, and scored as indicated in the rubric below. The student shall only be eligible for GA assessment if they met the subminima for both the demonstrations (D) and the tests (T).</td></tr>
<tr><td colspan="2">Student meets the above criteria</td><td colspan="2">Student fails to meet the above criteria</td></tr>
<tr><td>Pass</td><td>Marginal Pass</td><td>Marginal Fail</td><td>Fail</td></tr>
<tr><td>What is the consequence of unsatisfactory performance?</td><td colspan="4">If the candidate has not demonstrated the attribute, he/she cannot attain a pass mark for the module and has to repeat the module.</td></tr>
</table>

GA 4. Investigations, experiments and data analysis: Demonstrate competence to conduct investigations of complex engineering problems using research methods, including research-based knowledge, design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

<table>
<tr><td>How is the Attribute Assessed?</td><td colspan="4">This attribute is assessed by the evaluation of the student's ability to do fault finding using software, laboratory equipment, experiments and data analysis. Successful fault-finding depends fully on the development of a series of small, but very thorough, investigations which need to be planned carefully and executed competently. The GA assessment shall be based on the student’s final report (FR) as assessed by the lecturer. The reporting should demonstrate engagement with selected knowledge in the current research literature of the discipline. The awareness of the power of critical thinking must be demonstrated with different and creative approaches to evaluate emerging issues. This can be done by considering and commenting on the expected outcomes of different design and implementation solutions, ultimately selecting a final approach and providing reasons for doing so.
To be eligible for GA assessment, students will individually perform a set of demonstrations of the subsystems and a final demonstration of the complete system. Each demonstration is followed by a test to verify the student’s knowledge of the components, design principles and procedures. Average marks are computed for both demonstrations and tests.</td></tr>
<tr><td rowspan="3">What is Satisfactory Performance?</td><td colspan="4">Using the assessment material and opportunities, the student must show that he/she has designed and conducted investigations and experiments at a level consistent to that which a graduate would participate in an employment situation shortly after graduation. The student must show that he/she:
➢ Planned and conducted investigations and experiments;
➢ Conducted a literature search and critically evaluated material;
➢ Performed necessary analyses;
➢ Selected and used appropriate equipment or software;
➢ Analysed, interpreted and derived information from data;
➢ Drew conclusions based on evidence;
➢ Communicated the purpose, process and outcomes in a technical project report.
In order to achieve the outcome, the student’s final report (FR) is evaluated, as a whole, and scored as indicated in the rubric below. The student shall only be eligible for GA assessment if they met the subminima for both the demonstrations (D) and the tests (T).</td></tr>
<tr><td colspan="2">Student meets the above criteria.</td><td colspan="2">Student fails to meet the above criteria.</td></tr>
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GA 5. Engineering methods, skills and tools, including Information Technology: Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.

<table>
<tr><td>How is the Attribute Assessed?</td><td colspan="4">The project is structured in such a way that successful completion can only be achieved by competent use of information technology and a set of engineering skills and tools required to construct a system complying with the specifications. The GA assessment shall be based on the student's final report (FR) as assessed by the lecturer.
To be eligible for GA assessment, students will individually perform a set of demonstrations of the subsystems and a final demonstration of the complete system. Each demonstration is followed by a test to verify the student's knowledge of the components, design principles and procedures. Average marks are computed for both demonstrations and tests.</td></tr>
<tr><td rowspan="3">What is Satisfactory Performance?</td><td colspan="4">Using the assessment opportunities, the student must show that he/she used the engineering methods and the developing environment at a level consistent to that which a graduate would participate in an employment situation shortly after graduation. The student must show that he/she:
➢ Used methods, skills or tools effectively by appropriate selection, proper application and critical assessment of the results;
➢ Developed software applications using the IDE on a PC.
In order to achieve the outcome, the student's <u>final report (FR)</u> is evaluated, <u>as a whole</u>, and scored as indicated in the rubric below. The student shall only be eligible for GA assessment if they met the subminima for both the <u>demonstrations (D)</u> and the <u>tests (T)</u>.</td></tr>
<tr><td colspan="2">Student meets the above criteria.</td><td colspan="2">Student fails to meet the above criteria.</td></tr>
<tr><td>Pass</td><td>Marginal Pass</td><td>Marginal Fail</td><td>Fail</td></tr>
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</table>

GA 9. Independent learning ability: Demonstrate competence to engage in independent learning through well-developed learning skills.

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<tr><td>How is the Attribute Assessed?</td><td colspan="4">Throughout the module, students are required to interpret the specifications for the various subsystems and the final system, to study and interpret datasheets, application notes, and user manuals, to achieve the specifications, with little help from the lecturer. The GA assessment shall be based on the student's final report (FR) as assessed by the lecturer.
To be eligible for GA assessment, students will individually perform a set of demonstrations of the subsystems and a final demonstration of the complete system. Each demonstration is followed by a test to verify the student's knowledge of the components, design principles and procedures. Average marks are computed for both demonstrations and tests.</td></tr>
<tr><td rowspan="3">What is Satisfactory Performance?</td><td colspan="4">Using the assessment opportunities, the student must show that he/she has developed the ability to acquire knowledge in an independent fashion, apply such knowledge, and take responsibility for learning requirements. The student must show that he/she can:
➢ Reflect on own learning and determine learning requirements and strategies;
➢ Source and evaluate information;
➢ Access, comprehend and apply knowledge acquired outside formal instruction;
➢ Critically challenge assumptions and embraces new thinking.
In order to achieve the outcome, the student's <u>final report (FR)</u> is evaluated, <u>as a whole</u>, and scored as indicated in the rubric below. The student shall only be eligible for GA assessment if they met the subminima for both the <u>demonstrations (D)</u> and the <u>tests (T)</u>.</td></tr>
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</table>

8 Other Module Specific Information

8.1 Demonstrations

Practical demonstrations are evaluated during the practical periods. A student who misses an evaluation without permission will forfeit the marks for that evaluation. You must be ready to demonstrate **one hour after the start** of the practical session on the day on which your practical slot is scheduled.

The following additional concessions will apply for demonstrations:

- If a catastrophic hardware failure, on the day of your demonstration, prevents you from demonstrating during your slot, you may apply for the opportunity to demonstrate on the following Monday (09:00-10:00) – a mark penalty of 10% will automatically be applied. You must apply for the opportunity before the end of the specific demonstration time.
- With a valid absence permission, you must demonstrate on the first day back at university.

8.2 Tests

Tests are written individually during lecture periods and exam time. A student who misses a test will forfeit the marks for that test. All tests are open book.

8.3 Own work

The following rules will be applied rigorously:

- Each student must submit an individual project report containing his/her own writing in the indicated sections.
- Each student must build and debug his/her own hardware and may only demonstrate his/her own hardware and software.
- Marks are allocated individually for each demonstration.
- An individual may not acquire flow charts, pseudo code, programs or program fragments (unless it is in the public domain) from any other student prior to the demonstration. The demonstrated software must be the own work of the student.
- Software copying will be treated as plagiarism. Every individual must complete their own calculations. Component calculations, Spice models, Matlab M-files, Excel files, etc. must be individual work. Software development will be monitored using the GIT upload history. This rule also applies to design tasks and diagrams.
- After a demonstration, the student may obtain software from other sources, to replace code required for that demonstration. The original authors' names must appear in the commentary block of each function or procedure.
- Note: Do not leave your code on the laboratory computer hard drive – you are responsible for the security of your code.
- Plagiarism will be dealt with as per the engineering faculty policies and can lead to suspension from the university if found guilty.